



Project Report

on

Aerodynamics of shuttlecock

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What is aerodynamics ?

Aerodynamics is the branch of physics that studies the motion of air (or other gases) and the forces acting on objects moving through it. It focuses on how air flows around an object and how this flow influences forces like **drag**, **lift**, **thrust**, and **weight**. The primary goal of aerodynamics is to predict and optimize the performance, stability, and efficiency of moving bodies in air.

Principles:

Aerodynamics relies on concepts from fluid dynamics, including:

Aerospace Engineering

- Design of aircraft, spacecraft, and drones for improved fuel efficiency and stability.
- Analysis of lift and drag to ensure safe takeoff, flight, and landing.

Automotive Industry

- Streamlining car designs to reduce drag and improve fuel economy.
- Enhancing vehicle stability at high speeds.

Sports Engineering

- Optimizing shapes of balls, bicycles, helmets, and racetrack designs to reduce drag.
- Design of sports equipment like shuttlecocks, javelins, and racing suits.

Architecture

- Designing buildings and bridges to withstand wind loads and prevent unwanted vibrations.
- Improving natural ventilation through aerodynamic shaping.

Environmental and Energy Systems

- Wind turbine blade design for efficient energy capture.
- Predicting and controlling pollutant dispersion in urban areas.

Relevance to this project:

In the case of the badminton shuttlecock, aerodynamics explains why its unique shape ensures high drag and stability, causing it to always fall nose-first and follow a predictable trajectory.

Why is the shuttlecock special ?

The shuttlecock is a unique projectile in sports because of its distinctive aerodynamic behavior. Unlike balls used in games like tennis or cricket, the shuttlecock has an open conical skirt with feathers that greatly influence the airflow around it.

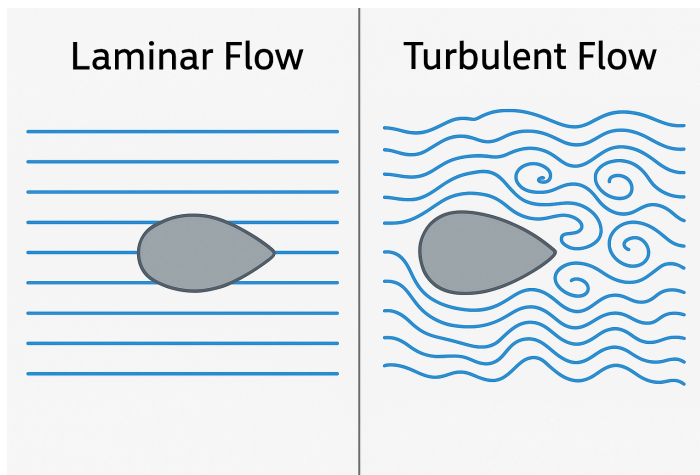
When the shuttlecock travels through air, it experiences two main types of fluid flow:

I. Laminar flow

This is a smooth and orderly movement of air in parallel layers with minimal mixing between them. In laminar flow, the fluid moves in a predictable path, resulting in less drag. Laminar flow usually occurs at low speeds or in streamlined shapes.

II. Turbulent flow

This is a chaotic and irregular movement of air with eddies, swirls, and rapid changes in velocity. Turbulent flow creates more drag but also helps stabilize certain objects in flight. It occurs at higher speeds or when the shape of an object disturbs airflow.



For a shuttlecock:

- At the start of its flight (just after being hit), the speed is high and the airflow quickly becomes **turbulent**, creating large drag forces that slow it down rapidly.
- As the shuttlecock slows, the flow transitions toward **laminar**, allowing it to descend more steadily.
- The conical feathered design ensures that no matter how it is struck, the shuttlecock turns nose-first into the direction of motion, maintaining stability and a predictable flight path.

This unique aerodynamic behavior of rapid deceleration due to turbulence followed by stable descent due to laminar flow makes the shuttlecock the slowest moving projectile in racket sports after a short distance, yet highly accurate in placement.

Aerodynamic characteristics of a badminton shuttlecock

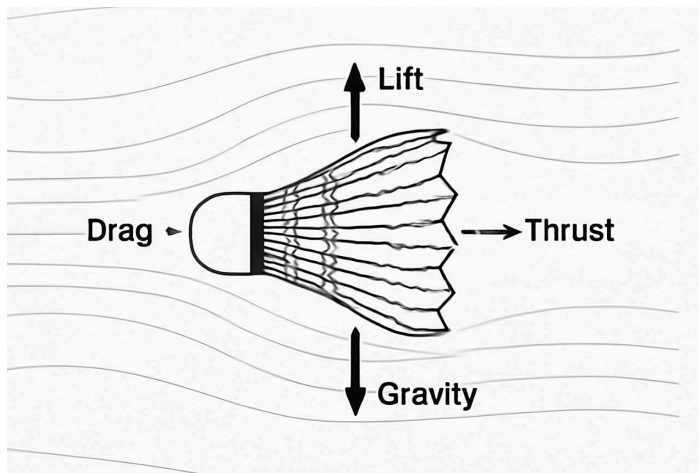
1. Reynolds Number (Re)

- A dimensionless number that predicts whether the flow around the shuttlecock is laminar or turbulent.

- Formula:

$$\text{Re} = \frac{\rho V D}{\mu}$$

- ρ : Air density
 - V : Velocity of shuttlecock
 - D : Characteristic length (diameter)
 - μ : Dynamic viscosity of air
- Shuttlecocks generally have **low-to-moderate Re** due to their small size and high drag, causing turbulent wake behind them.



2. Drag Force (F_D)

- The resistive force opposes motion through air.
- Formula:

$$F_D = \frac{1}{2} C_d \rho A V^2$$

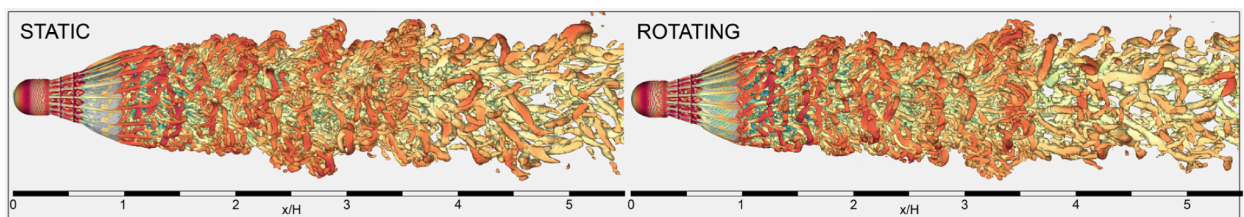
- C_d : Drag coefficient (high for shuttlecocks, ~0.7–1.0)

- **A**: Cross-sectional area
 - **V**: speed of shuttlecock
 - ρ : Air density
- The conical skirt increases drag, slowing the shuttlecock rapidly for better control.

3. Wake Formation and Flow Separation

$$\Delta P = P_{\text{front}} - P_{\text{rear}} \rightarrow F_D = \Delta P \cdot A$$

- As air flows around the shuttlecock, it detaches from the surface, creating **flow separation**.
- This produces a **low-pressure turbulent wake** behind the shuttlecock.
- The skirt's gaps help stabilize the wake, keeping the shuttlecock **nose-first** during flight.



4. Strouhal Number (St)

- Dimensionless number describing vortex shedding frequency in the wake.

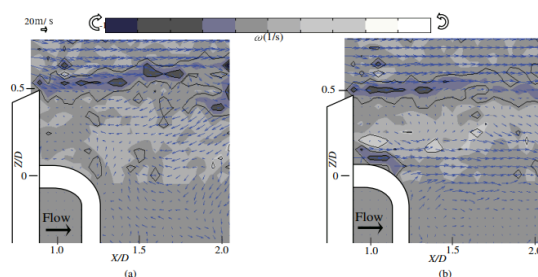


Fig.7 Density map of vorticity in X-Z plan for $Re=8.6 \times 10^4$ ($\alpha=0^\circ$) for (a) shuttlecock without gap; (b) ordinary shuttlecock.

- Formula:

$$St = \frac{fL}{V}$$

- **f**: Vortex shedding frequency
 - **L**: characteristic length (eq: skirt diameter)
- For shuttlecocks, St helps predict the oscillations and size of the wake, affecting stability.

5. Lift Force (L)

- Formula

$$L = \frac{1}{2} \rho V^2 A C_L$$

- ρ : Air density
 - **V**: velocity of shuttlecock
 - **A**: Cross sectional of air
 - **C_L**: Lift coefficient
- Caused by slight asymmetries or spin, but minimal compared to drag in shuttlecocks.
- Any generated lift helps maintain **trajectory stability** rather than increasing range.

6. Terminal Velocity (V_t)

- The constant speed reached when **drag force equals gravitational force**.
- Formula:

$$V_t = \sqrt{\frac{2mg}{CD\rho A}}$$

- **m**: Mass of shuttlecock
 - **g**: Acceleration due to gravity
 - **A**: cross sectional area
 - **C_D**: drag coefficient

- ρ : Air density
- Shuttlecocks have low terminal velocity (~6–7 m/s) due to high drag, enabling controlled, short-distance play.

Mass Conservation (Continuity Equation)

Definition:

Mass conservation means that the total mass of air in a given region of flow always remains constant — no air is created or lost in the process.

Equation:

$$\nabla \cdot \mathbf{u} = 0$$

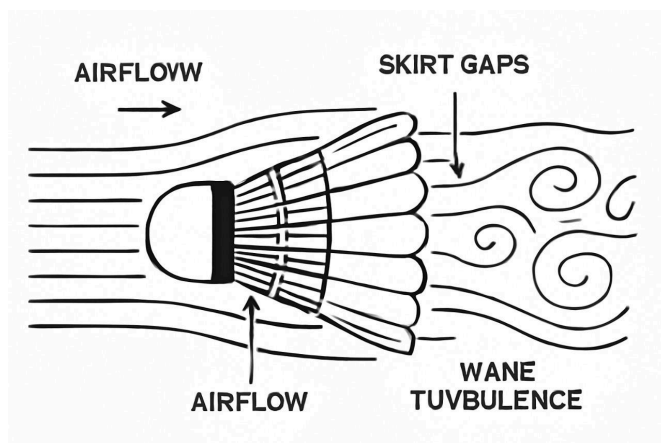
where $\mathbf{u}$ represents the velocity vector of the airflow. This is valid for incompressible flows, like the subsonic air movement around a shuttlecock.

Meaning in Simple Terms:

Whatever amount of air flows into a space must be exactly equal to the amount that flows out.

In Shuttlecock Flight:

- Air travels not only around the shuttlecock's skirt but also through the gaps in the skirt material.
- This "bleed-through" flow changes the wake (the turbulent air zone) behind the shuttlecock.
- Altering the wake pattern directly impacts drag (slowing force) and flight stability — helping the shuttlecock maintain a smooth and predictable path.



Momentum Conservation (Navier–Stokes Equations)

Definition:

Momentum conservation explains how the movement of air changes when it is acted upon by forces such as pressure gradients, viscosity (fluid friction), and gravity.

In Shuttlecock Aerodynamics:

- These principles govern key aerodynamic effects like flow separation, vortex shedding, and turbulence formation.
- They also describe how pressure is distributed across the shuttlecock's surface during flight.

Role of Skirt Gaps:

- The gaps in the skirt act like tiny nozzles, releasing thin jets of air.
- These air jets disturb and rebalance the pressure around the shuttlecock.
- This controlled disturbance helps maintain the cork-first orientation, even under fast and powerful hits.

Use in Analysis:

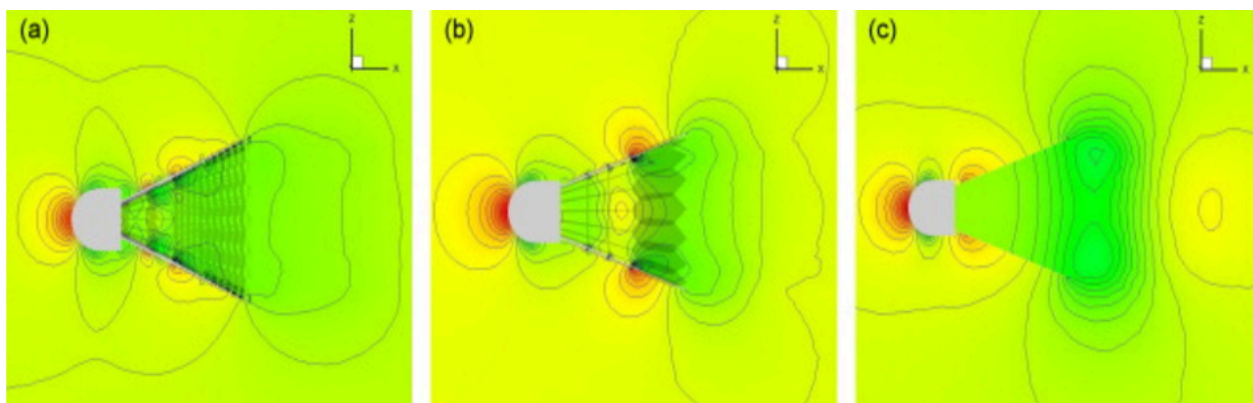
- Engineers and researchers use Computational Fluid Dynamics (CFD) simulations based on the Navier–Stokes equations to map airflow patterns.
- These simulations help visualize the wake size, measure drag, and understand how design changes influence stability and performance.

Why It Matters:

The shuttlecock's flight is not accidental, it's the result of precise aerodynamic balancing. Mass conservation ensures smooth and continuous airflow, while momentum conservation explains how forces interact to keep the shuttlecock stable. Together, these principles make the shuttlecock slow down quickly yet remain perfectly aligned in flight, making badminton both challenging and predictable.

Analysis of Pressure Effects on Different Types of Shuttlecocks

This section examines the variations in pressure distribution for three types of shuttlecocks: feather, plastic, and solid using both experimental observations from the pressure graphs and theoretical aerodynamic analysis.



1. Feather Shuttlecock

- The pressure graph for the feather shuttlecock shows significant fluctuations, with large peaks and troughs. This high variation arises due to the gaps between the feathers, which disrupt the airflow and induce rotational motion.
- From a fluid dynamics perspective, the gaps act as periodic discontinuities in the flow field, similar to singularities in potential flow theory. These discontinuities lead to vortex shedding, which appears in the complex potential function $F(z)=\phi+i\psi$ as branch cuts.
- The resulting flow separation creates regions of low pressure behind each gap, producing a highly variable pressure coefficient (C_p) along the shuttlecock's surface.
- This large pressure variation, although contributing to higher drag, generates stabilizing spin.
- The rotational stability ensures that feather shuttlecocks maintain straighter and more controlled flight trajectories despite reduced aerodynamic efficiency.

2. Plastic Shuttlecock

- The plastic shuttlecock displays smaller and smoother pressure variations in its graph. While some oscillations remain, they are less extreme compared to the feather shuttlecock. The continuous synthetic vanes present fewer interruptions to the airflow, reducing turbulence.
- Mathematically, the regular vane pattern can be represented through conformal mapping techniques. The C_p distribution here is more uniform, with moderately negative values on the leeward side due to predictable flow separation at each vane edge. This results in lower drag and faster flight speeds.
- However, the reduced pressure variation produces less stabilizing spin, making plastic shuttlecocks more prone to wobbling and less precise in flight than feather ones.

3. Solid Shuttlecock

- The solid shuttlecock exhibits an almost perfectly smooth pressure graph, indicating minimal variation in C_p .
- This case resembles an idealized potential flow problem around a hemispherical body, where the complex potential $F(z)$ remains analytic over the entire domain.
- Here, the C_p distribution follows ideal predictions of maximum pressure at the stagnation point, gradually decreasing toward the sides with no sudden drops due to flow separation.
- This configuration minimizes drag but eliminates any vorticity-generating effects, resulting in no stabilizing spin.
- Consequently, while theoretically efficient in terms of aerodynamics, the solid shuttlecock would be unstable and difficult to control in actual gameplay.

Comparative Analysis of Shuttlecock Types

A comparison of key aerodynamic parameters for feather, plastic, and solid (gapless) shuttlecocks reveals clear performance distinctions:

- Spin:
Feather shuttlecocks generate strong natural spin due to disrupted airflow through feather gaps, enhancing stability. Plastic shuttlecocks spin more weakly due to fewer



gaps, making them more prone to wobble. Solid shuttlecocks exhibit no spin, causing immediate directional changes when disturbed.

- **Flight Path:**

Feather shuttlecocks maintain a stable, parabolic trajectory. Plastic variants show occasional wobble, while solid ones deviate unpredictably from their intended path.

- **Drag:**

Drag is highest in feather shuttlecocks due to irregular edges and gaps, moderate in plastic due to partial smoothness, and lowest in solid shuttlecocks owing to a fully smooth surface.

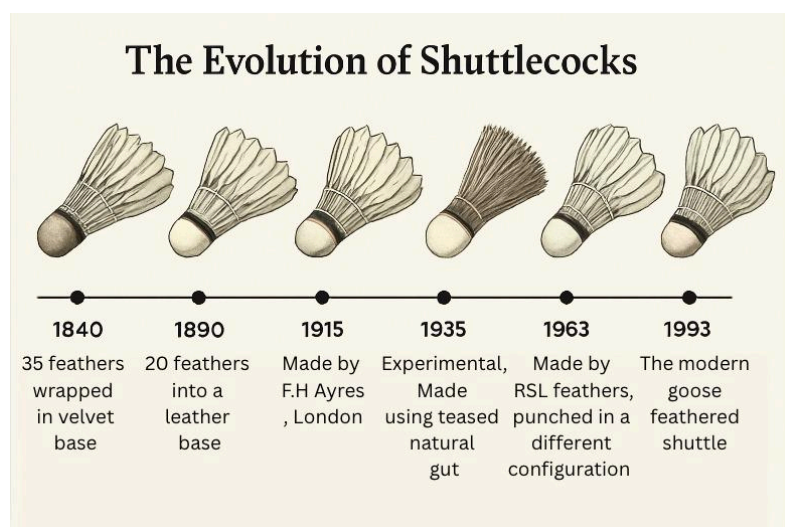
- **Velocity:**

Higher drag in feather shuttlecocks results in the lowest flight speed, whereas plastic types achieve moderate speeds, and solid shuttlecocks attain the highest velocity due to minimal resistance.

- **Vortices:**

Feather shuttlecocks produce strong trailing vortices that act as stabilizing fins. Plastic versions form weaker vortices, offering limited stability, while solid shuttlecocks generate no vortices, resulting in poor control.

Timeline: Shuttlecock Evolution (1840–1993)



1840 – Early Feather Shuttlecocks

Structure:

~35 feathers inserted into a velvet-wrapped base.

Technical terms:

Feathers: Natural goose or duck feathers—light and aerodynamic.

Velvet-wrapped base: A soft exterior around the base, likely over cork or leather.

Pros:

- Used readily available natural materials.
- Provided decent flight stability for early play.

Cons:

- Heavy and bulky; hard to control.
- Low precision; flight path inconsistent.

1890 – Leather Base Designs

Structure:

~20 feathers inserted into a leather base.

Pros:

- Leather is more robust than velvet—offering improved durability.
- Slight weight reduction by using fewer feathers and sturdy base.

Cons:

- Still heavy and expensive to produce.
- Flight quality still varied significantly.

1915 – F. H. Ayres, London

Structure:

Likely improved craftsmanship by a specific maker—better feather alignment and tighter base integration.

Pros:

- Enhanced consistency in shape and flight.
- Possibly lighter and better weighted for control.

Cons:

- Artisanal production kept costs high.
- Still based on natural materials—fragile and variable.

1935 – Experimental Shuttlecocks with Natural Gut

Structure:

Feather shuttlecock incorporating teased natural gut.

Technical Terms:

Natural gut: Animal-derived fiber (like lamb intestine), known for elasticity and tensile strength.

Pros:

- Added structural reinforcement around feathers potentially improved shape retention and flight stability.

Cons:

- The gut is hygroscopic—affected by moisture.
- Increased fragility and manufacturing complexity.

1963 – RSL Shuttle with Triple Stitching

Structure:

Feather shuttlecocks manufactured by RSL (Reinforced Shuttlecocks Ltd), using three rows of stitching in base assembly.

Pros:

- Stitching increased durability and kept feathers in place longer.
- Better consistency in mass production.

Cons:

- Still fragile compared to synthetic alternatives.
- Higher production cost due to intricate stitching.

1993 – Modern Goose-Feather Shuttlecock

Structure:

Contemporary standard: ~16 goose feathers embedded in a composite cork-leather base.

Technical Terms:

Cork base: Lightweight, elastic natural material providing rebound.

Isometric feather arrangements: Often standardized for flight precision.

Pros:

- Lightweight (≈ 4.75 – 5.50 g), aerodynamic, and used in competition
- Highly precise, with consistent flight and control—favored in tournaments.

Cons:

Very fragile—feathers break after a few games

Sensitive to humidity and require humidification before use

Evolution of Badminton Racquets (1909–2024)



1. 1909 – Wooden Era

Material:

Solid hardwood (e.g., ash, hickory).

Weight:

~150–200 g.

Strings:

Natural gut (animal intestine fibers).

Technical terms:

Natural gut: Highly elastic, good feel, but loses tension quickly and is sensitive to humidity.

Pros:

- Good control and touch.
- Durable against minor impacts.

Cons:

- Heavy → slower swings.
- Warps with moisture.

- Low tension capacity (~15–18 lbs).

2. 1924 – Wood + Steel Joint

Material:

Wood frame + steel T-joint (metal connector between shaft and head).

Weight:

~140–180 g.

Technical terms:

T-joint: Junction that stabilizes the racquet head and prevents twisting.

Pros:

- Better durability and head stability.
- Less frame warping.

Cons:

- Still heavy.
- Slightly head-heavy → tiring in long rallies.

3. 1930 – Split-T Frame

Material:

Wood frame with split T metal connector.

Technical terms:

Split T: T-joint split into two branches for extra head support.

Pros:

- More consistent string bed tension.
- Slight improvement in accuracy.

Cons:

- Weight reduction minimal.
- Still prone to weather damage.

4. 1965 – Wood + Aluminum Head

Material:

Wooden handle/shaft + aluminum alloy head.

Weight:

~120–150 g.

Technical terms:

Aluminum alloy: Lightweight metal blend, more rigid than wood.

Pros:

- Lighter → faster swings.
- More resistant to warping.

Cons:

- Can dent if hit against hard surfaces.
- Balance is sometimes uneven due to mixed materials.

5. 1966 – Full Metal Frame

Material:

Entirely aluminum or steel.

Weight:

~110–140 g.

Technical terms:

Nylon strings: Synthetic polymer strings, stronger and weather-resistant.

Pros:

- Higher string tension possible (~20–22 lbs).
- Much more durable than wood.

Cons:

- Feels “stiff” with less shuttle feedback.
- Heavier than later carbon frames.

6. 1973 – Laminated Wood

Material:

Layers of wood bonded together with adhesives.

Technical terms:

Lamination: Layering wood for added strength without adding much weight.

Pros:

- Cheaper than metal.
- Retained traditional feel.

Cons:

- Still heavier than metal.
- Lower smash speed potential.

7. 1983 – Graphite Composites

Material:

Graphite (carbon) + resin + sometimes fiberglass.

Weight:

~90–110 g.

Technical terms:

Graphite: Carbon form with high stiffness-to-weight ratio.

Composite: Blend of materials for optimal properties.

Pros:

- Extremely light but stiff.
- Supports higher tension (~24–28 lbs).
- Faster swing → higher shuttle speed.

Cons:

- More expensive.
- Brittle under extreme impact.
-

8. 2014 – High-Modulus Graphite + Nanotech

Material:

High-modulus graphite (stiffer carbon) + nanomaterial reinforcement.

Weight:

~80–95 g.

Technical terms:

High modulus: Higher stiffness without adding weight → better energy transfer.

Nanotech: Nano-scale particles to improve strength and shock absorption.

Pros:

- Huge sweet spot with isometric head shape.
- Can handle 30–35 lbs tension for aggressive play.
- Very aerodynamic frame.

Cons:

- Requires skilled handling to use high tension effectively.
- Can feel less forgiving to beginners.

9. 2024 – Ultra-Modern Carbon + Smart Tech

Material:

High-modulus graphite + nanomaterials + tungsten grommets + sometimes built-in sensors.

Weight:

~75–85 g.

Technical terms:

Tungsten grommets: Harder inserts where strings pass through, reducing friction and increasing tension lifespan.

Smart sensors: Track swing speed, impact location, smash force.

Pros:

- Fastest smashes (>400 km/h possible at pro level).
- Adjustable balance points for offense or defense.
- Extremely light and maneuverable.

Cons:

- Very expensive.
- Thin frame → more fragile under mishits.
- High performance depends heavily on player skill.

Conclusion

- The shuttlecock's aerodynamic behavior is defined by high drag, flow separation, and stabilizing wake patterns.
- Its conical skirt and feather gaps cause rapid deceleration while ensuring a stable, nose-forward flight.
- Analysis of Reynolds number, drag force, Strouhal number, and terminal velocity shows that the design prioritizes control over speed.
- Feather shuttlecocks provide the best stability and precision, while plastic and solid types trade stability for durability or speed.
- Mass and momentum conservation principles explain how airflow interacts with shuttlecock geometry to maintain flight stability.
- The evolution of shuttlecocks and racquets reflects ongoing innovation to improve durability, precision, and player performance.
- The shuttlecock's simple form conceals a highly optimized aerodynamic design, offering lessons for sports engineering and aerodynamic applications.

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